

Shengkai Tao

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EDUCATION

University of North Carolina at Chapel Hill

Aug. 2023 – Now

B.S. Statistics and Data Science, B.A. Sociology

Chapel Hill, NC

- **Courses:** Data Structures, Algorithms & Analysis, Probability, Optimization, Regression analysis, Modeling, Sociology Theories.
- **Honors:** Dean's List for All Semesters

TECHNICAL SKILLS

Languages: Python (pandas, plotly, scipy, Selenium, PyTorch), SQL, R, Java, C++

Tools: Tableau, Power BI, LaTeX, Microsoft Power Automate, Excel, Docker

Data Science: A/B testing, predictive modeling, Causal inference, Regression analysis, Random forests

AI Engineering: Multi-Agent Systems, Google's Agent Development Kit (ADK), Prompt Engineering

Workflow & Collaboration: Git, Jupyter, VSCode, Agile development

Optimization & Measurement: Experience with designing and analyzing experiments, modeling, and performance evaluation using statistical metrics

EXPERIENCE

AI Agent Engineer Intern

May. 2025 – Now

Rowdy Braude

Remote, base in Charlotte, NC

- Developed and deployed a multi-agent system using **Python** and **Google's Agent Development Kit (ADK)** to conduct comprehensive strategic analysis of pharmaceutical brands. Aimed for replacing job now heavily relied on human.
- Engineered an AI-driven workflow to automate the generation of creative briefs and concepts, transforming complex clinical data into compelling, optimized promotional messages, including posters and images that can be used for advertising immediately.
- Designed and implemented a modular architecture with specialized AI agents for **market positioning analysis**, **competitive landscape assessment**, and **target audience evaluation**, incorporating behavioral science and cultural anthropology insights.
- Utilized an AI critique loop to refine and **optimize marketing messages** against best practices, customer needs, and competitive differentiation, resulting in a **portfolio of data-informed, rigorously critiqued creative concepts**.
- Integrated custom tools with agents for enhanced functionality, including web scraping, report generation and data formatting, and utilized **Docker** for containerization and deployment.

Compliance Consultant Intern

May. 2024 – Aug. 2024

Ernst & Young

Shanghai, China

- Contributed to risk consulting engagements for multilateral development banks and non-bank financial institutions, with a focus on regulatory compliance, internal controls, and organizational risk mitigation.
- Conducted in-depth walkthroughs and control assessments across **4 major departments**, uncovering **10+ key risk points** and identifying critical compliance and operational inefficiencies.
- Leveraged **Python** and **Excel** to evaluate internal control effectiveness, perform risk scoring, and uncover areas for process optimization.
- Delivered **actionable recommendations** to improve enterprise risk frameworks and ensure alignment with international regulatory standards, directly supporting client readiness for cross-border operations.
- Used **Python pandas** to analyze audit datasets, identify high-risk patterns, and support data-informed decision-making in audit planning and reporting.
- Co-authored comprehensive audit reports with a focus on **accuracy, clarity, and practical value**, enhancing transparency and stakeholder engagement.

PROJECTS

ASA DataFest 2025 | Best Visualization Award, Office Expansion Analysis

Apr. 2025

- Won **Best Visualization Award** out of 58 teams at ASA DataFest 2025 (Duke University) for producing actionable and intuitive graphics that shaped client-facing business strategy.
- Collaborated in a cross-functional team of 5 to analyze **U.S. commercial real estate transactions (2018-2024)**, targeting high-growth regions for office expansion in the tech, legal, and financial sectors.
- Developed **treemaps** and **stacked bar charts** using **Python (matplotlib, plotly)** to highlight the emergence of Miami-Dade County as a legal and financial hub.
- Engineered a **Random Forest classifier** to recommend optimal office locations based on rent, square footage, leasing trends, and transaction type, achieving **92.5% prediction accuracy**.

Marketing Campaign Optimization | Real World Simulation Project

Jan. 2024 - May. 2024

- Developed a machine learning model to **predict customer responsiveness to marketing campaigns for iFood**, leveraging customer demographics, spending habits, and platform usage data.
- Conducted feature engineering and exploratory data analysis using Python (**pandas, numpy, matplotlib**), identifying key predictors such as past campaign acceptance, customer tenure, and total spending.
- Implemented and evaluated **Decision Tree models**, comparing a feature-selected "Small Tree Model" with a full-featured "Large Tree Model," **both achieving 88% accuracy**.
- Addressed data biases, ensured compliance with **Brazil's LGPD data** regulations, and proposed deployment strategies for targeted marketing segmentation.